

REC Data Model Specification

This document details the data structures for on-chain state assets in Hyperledger Fabric and off-chain relational models in PostgreSQL.

1. On-Chain Data Model: RECAsset

The on-chain state is stored as a serialized JSON object in the Fabric World State (LevelDB/CouchDB) under the key recId.

```
{
  "recId": "REC-000001",
  "generatorId": "GEN-SOLAR-01",
  "energySource": "SOLAR",
  "generationDate": "2026-09-10",
  "generationMWh": 100,
  "issuedQuantity": 100,
  "currentOwner": "ISSUER-ORG",
  "activeQuantity": 100,
  "retiredQuantity": 0,
  "status": "ACTIVE",
  "documentHash": "a591a6d40bf420404a011733cfb7b190d62c65bf0bcda32b57b277d9ad9f146e",
  "createdAt": "2026-09-12T19:27:25.548Z",
  "updatedAt": "2026-09-12T19:27:31.125Z",
  "balances": {
    "ISSUER-ORG": 60,
    "CORP-BUYER-A": 40
  }
}
```

1.1 Field Definitions

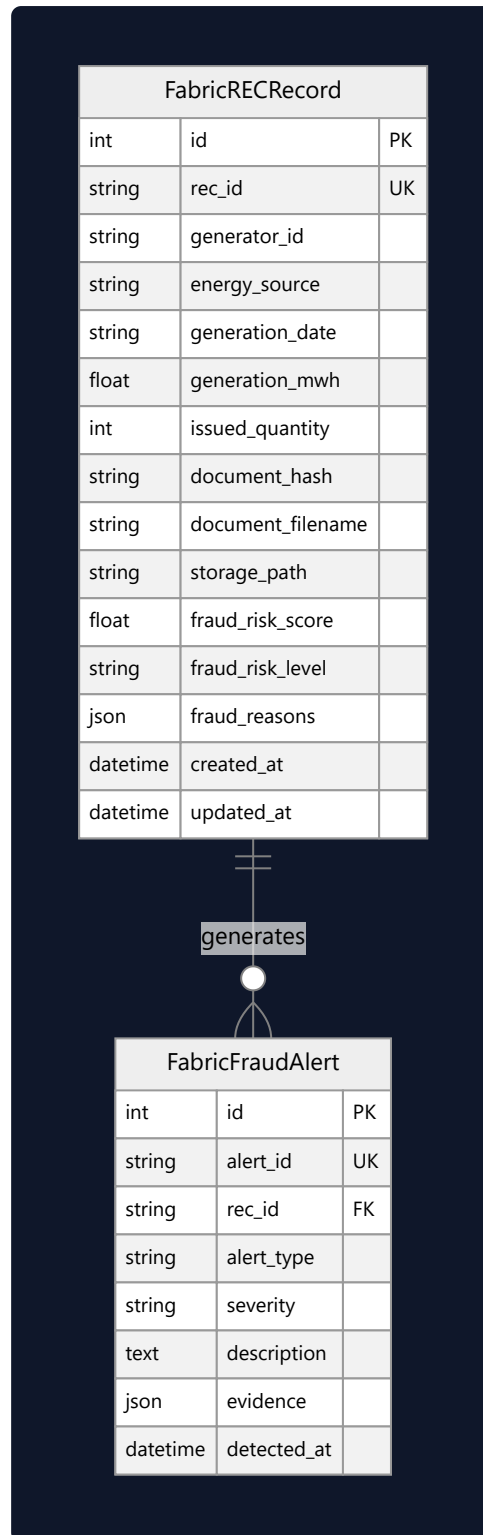
Field	Type	Description
recId	string	Unique immutable primary key (e.g., REC-000001).
generatorId	string	Power generation facility identifier (e.g., GEN-001).
energySource	string	Fuel technology type: SOLAR , WIND , HYDRO , BIOMASS .
generationDate	string	Generation vintage period (YYYY-MM-DD).
generationMWh	number	Total megawatt-hours of verified clean electricity.
issuedQuantity	number	Total certificate units minted (1 REC = 1 MWh).
currentOwner	string	Primary custodian or latest transferring participant account.
activeQuantity	number	Available active units eligible for market trading or retirement.
retiredQuantity	number	Units permanently redeemed for carbon accounting / Scope 2.
status	string	State flag: ACTIVE , RETIRED , CANCELLED .
documentHash	string	64-character SHA-256 cryptographic digest of proof document.
createdAt	string	ISO-8601 creation timestamp recorded by peer transaction proposal.
updatedAt	string	ISO-8601 timestamp of most recent state modification.
balances	object	Key-value mapping of { [ownerAccount]: quantity } maintaining sub-allocations.

1.2 Invariant Conservation Laws

- $\text{activeQuantity} + \text{retiredQuantity} \equiv \text{issuedQuantity}$
- $\sum_{\text{owner}} \text{balances}[\text{owner}] \equiv \text{activeQuantity}$
- For all owners: $\text{balances}[\text{owner}] \geq 0$
- If $\text{status} == \text{"CANCELLED"}$, no balance transfers or retirements are permitted.
- If $\text{activeQuantity} == 0$, status is irrevocably **RETIRED**.

2. Off-Chain Relational Data Model (PostgreSQL)

Located in `backend/app/models/fabric_rec.py` .



2.1 Table: **fabric_rec_records**

Maintains off-chain index of all issued RECs, storing file paths to off-chain documents and caching forensic fraud scores computed by the Fraud Detection Engine.

2.2 Table: **fabric_fraud_alerts**

Stores real-time forensic detection events (e.g. duplicate document hashes, suspicious transfer velocity, circular trading patterns).

